

Guideline Title: Allergic Reaction/Anaphylaxis

Guideline Number: GUID-3203-M004

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Revision dates: 4/30/24	Developed by: Air Care Team
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Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** Recognition of a patient presenting with anaphylaxis; provide and maintain adequate oxygenation, ventilation, and end-organ tissue perfusion; and to provide emergent care and expeditious transport to the nearest appropriate medical facility.
- II. **DEFINITIONS:**
 - a. Mild reaction: itching/hives
 - b. Moderate reaction: dyspnea, wheezing, chest tightness, facial swelling
 - c. Severe Reaction: < 90 mmHg, MAP < 65, stridor, severe respiratory distress, vomiting/diarrhea
- III. **DEPARTMENT GUIDELINE:**
 - a. Establish care as defined by *GUID3203-G002 – General Patient Care*.
 - b. While assessing the patient, attempt to determine the etiology of the allergic reaction.
 - i. If medication infusion reaction suspected, stop the infusion.
 - c. In cases of severe distress, administer epinephrine:
 - i. **Epinephrine (1:1,000) 0.5 mg IM in the anterior thigh q 5-15 min (or sooner if rapidly developing shock)** up to 3 doses.
 1. Massage injection site vigorously for 30-60 seconds
 - d. Obtain large bore IV/IO access
 - e. If patient is hypotensive or displays signs of inadequate tissue perfusion, administer a fluid bolus of 250 ml of 0.9% Normal Saline. Repeat as necessary.
 - i. If patient remains hypotensive after IM Epinephrine and multiple fluid boluses, consider initiating an **Epinephrine (4mg/250ml NS or D5W) infusion at 0.1mcg/kg/min** and titrate every 2-3 min by 0.05mcg/kg/min. Maintenance rate 0.025-0.5 mcg/kg/min.
 1. If patient is on beta-blocker therapy and appears refractory to epinephrine treatment, consider **Glucagon, 1-5 mg slow IV bolus**
 2. For additional management refer to *GUID3203-M011 – Hypotension*.
 - f. Perform a rapid assessment of the patient's airway
 - i. Intubation should be performed if stridor or respiratory distress due to upper airway compromise, significant edema of tongue, oropharyngeal tissues, or voice alteration per *GUID3203-G008 - Rapid Sequence Induction for Endotracheal Intubation*.

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- ii. Emergency cricothyroidotomy can be considered to secure airway if severe upper airway edema prevents access to glottic opening with the use of a video laryngoscope per *GUID3203-PR007 – Cricothyrotomy – Surgical*.
 - g. If the patient is experiencing dyspnea (with or without wheezing), administer
 - i. High flow oxygen
 - ii. Inhaled bronchodilators and repeat as needed: **Albuterol 2.5mg/3ml NSS** and **Ipratropium Bromide 0.02% 0.5mg/2.5ml** via NEB per *GUID3203-PR012 - Inhalation Therapy*.
 - iii. Do not delay endotracheal intubation if patient has rapidly evolving airway obstruction.
 - h. Administer antihistamine
 - i. **Diphenhydramine 25-50 mg slow IV/IO/IM**
 - ii. Consider **Methylprednisolone 125 mg IV** when hemodynamically stable.
 - i. Consider H2 blocker if at sending facility: **Famotidine 20 mg PO or 0.25 mg/kg IV** over 2 minutes.
- IV. **DOCUMENTATION:**
- a. EMS Charts
- V. **REFERENCES:**
- a. Cheng, A., Canadian Paediatric Society, Acute Care Committee. (2011). Emergency treatment of anaphylaxis in infants and children. *Paediatr Child Health*, 16(1), 35-40.
 - b. Weingart, S. D. & Borshiff, D. C. (2018). *The Resuscitation Crisis Manual*. Leeuwin Press.